**Southern Oil Refinery 2**
**SOR2-MEC-CAL-C101_REV A**
**FRACTIONATION COLUMN**
**PV Elite 28  Licensee: SL User**
**FileName : C101 REV A----------**
**Nozzle Flange MAWP:  Step: 19  3:18pm Aug 19,2026**

### Nozzle Flange MAWP Results: (kPa & °C)

| Nozzle Description | Flange Rating Ope. | Flange Rating Ambient | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure - - - - UG-44 | Max Pressure 50% | Max Pressure DNV |
|---|---|---|---|---|---|---|---|---|---|
| B | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5138.7 | 429.4 | 3712.1 | 429.4 |
| M5 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L6 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L4 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L5 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| L3 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| T1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| P1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| N1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| A | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 509.0 | 3574.3 | 3712.1 | 3712.1 |
| T2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| M4 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| N2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| C | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5138.7 | 429.4 | 3712.1 | 429.4 |
| D | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5132.5 | 435.6 | 3712.1 | 435.6 |
| Q1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| T3 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| M3 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| Q2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| R1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| E | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5138.7 | 429.4 | 3712.1 | 429.4 |
| F | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5147.7 | 420.4 | 3712.1 | 420.4 |
| T4 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| M2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| R2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| G | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5138.7 | 429.4 | 3712.1 | 429.4 |
| H | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 3875.4 | 1692.7 | 3712.1 | 1692.7 |
| S1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| T5 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| M1 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| P2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| S2 | 800.0 | 1960.0 | 360 | 150 | GR 1.1 | ... | ... | ... | ... |
| J | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 5147.7 | 420.4 | 3712.1 | 420.4 |
| K | 3712.1 | 5100.0 | 360 | 300 | GR 1.1 | 3892.0 | 1676.0 | 3712.1 | 1676.0 |
| Min Rating | 420.4 | 1960.0, [for Core Elements] | | | | ... | ... | ... | ... |

**Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)**

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

| Nozzle Description | Gasket Dimension | Fm Value | Resolved Moment | Axial Force | MAWP | Maximum Pressure | Pass Fail |
|---|---|---|---|---|---|---|---|

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**FRACTIONATION COLUMN**
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**FileName : C101 REV A----------**
**Nozzle Flange MAWP:  Step:  19  3:18pm  Aug 19,2026**

| | G mm | | ME N-m | FE N | PD kPa | Pmax kPa | |
|---|---|---|---|---|---|---|---|
| B | 77.900 | 0.50 | 460 | 900 | 420.4 | 429.4 | Pass |
| A | 988.600 | 0.10 | 90188 | 25927 | 420.4 | 3574.3 | Pass |
| C | 77.900 | 0.50 | 460 | 900 | 420.4 | 429.4 | Pass |
| D | 138.176 | 0.50 | 2546 | 3300 | 420.4 | 435.6 | Pass |
| E | 77.900 | 0.50 | 460 | 900 | 420.4 | 429.4 | Pass |
| F | 196.088 | 0.50 | 7354 | 5500 | 420.4 | 420.4 | Pass |
| G | 77.900 | 0.50 | 460 | 900 | 420.4 | 429.4 | Pass |
| H | 248.539 | 0.50 | 11314 | 6000 | 420.4 | 1692.7 | Pass |
| J | 196.088 | 0.50 | 7354 | 5500 | 420.4 | 420.4 | Pass |
| K | 357.251 | 0.50 | 33517 | 15000 | 420.4 | 1676.0 | Pass |

Evaluating Flange on Nozzle [B]:
```
  = 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
  = 16(460)+4*abs(900)77.900 <= pi*77.900^3[(3712.053-420.361)+.500*3712.053]
  = 0.76344E+04 <= 0.76479E+04  Check Passes
```

Maximum Pressure for Flange on Nozzle B [Pmax]:
```
  = min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
  = min( 3712.053, 0.500 * 3712.053 + 3712.053 - 7634.4/(pi * 77.900^3) )
  = min( 3712.053, 429.409 )
  = 429.409 kPa
```

Evaluating Flange on Nozzle [A]:
```
  = 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
  = 16(90188)+4*abs(25927)988.600 <= pi*988.600^3[(3712.053-420.361)+.100*3712.053]
  = 0.15456E+07 <= 0.11122E+08  Check Passes
```

Maximum Pressure for Flange on Nozzle A [Pmax]:
```
  = min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
  = min( 3712.053, 0.100 * 3712.053 + 3712.053 - 1545574.4/(pi * 988.600^3) )
  = min( 3712.053, 3574.263 )
  = 3574.263 kPa
```

Evaluating Flange on Nozzle [C]:
```
  = 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
  = 16(460)+4*abs(900)77.900 <= pi*77.900^3[(3712.053-420.361)+.500*3712.053]
  = 0.76344E+04 <= 0.76479E+04  Check Passes
```

Maximum Pressure for Flange on Nozzle C [Pmax]:
```
  = min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
  = min( 3712.053, 0.500 * 3712.053 + 3712.053 - 7634.4/(pi * 77.900^3) )
  = min( 3712.053, 429.409 )
  = 429.409 kPa
```

Evaluating Flange on Nozzle [D]:
```
  = 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
  = 16(2546)+4*abs(3300)138.176 <= pi*138.176^3[(3712.053-420.361)+.500*3712.053]
  = 0.42554E+05 <= 0.42680E+05  Check Passes
```

Maximum Pressure for Flange on Nozzle D [Pmax]:
```
  = min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
  = min( 3712.053, 0.500 * 3712.053 + 3712.053 - 42554.0/(pi * 138.176^3) )
  = min( 3712.053, 435.584 )
  = 435.584 kPa
```

Evaluating Flange on Nozzle [E]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(460)+4*abs(900)77.900 <= pi*77.900^3[(3712.053-420.361)+.500*3712.053]
= 0.76344E+04 <= 0.76479E+04  Check Passes
```

Maximum Pressure for Flange on Nozzle E [Pmax]:
```
= min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
= min( 3712.053, 0.500 * 3712.053 + 3712.053 - 7634.4/(pi * 77.900^3) )
= min( 3712.053, 429.409 )
= 429.409 kPa
```

Evaluating Flange on Nozzle [F]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(7354)+4*abs(5500)196.088 <= pi*196.088^3[(3712.053-420.361)+.500*3712.053]
= 0.12198E+06 <= 0.12198E+06  Check Passes
```

Maximum Pressure for Flange on Nozzle F [Pmax]:
```
= min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
= min( 3712.053, 0.500 * 3712.053 + 3712.053 - 121978.2/(pi * 196.088^3) )
= min( 3712.053, 420.361 )
= 420.361 kPa
```

Evaluating Flange on Nozzle [G]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(460)+4*abs(900)77.900 <= pi*77.900^3[(3712.053-420.361)+.500*3712.053]
= 0.76344E+04 <= 0.76479E+04  Check Passes
```

Maximum Pressure for Flange on Nozzle G [Pmax]:
```
= min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
= min( 3712.053, 0.500 * 3712.053 + 3712.053 - 7634.4/(pi * 77.900^3) )
= min( 3712.053, 429.409 )
= 429.409 kPa
```

Evaluating Flange on Nozzle [H]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(11314)+4*abs(6000)248.539 <= pi*248.539^3[(3712.053-420.361)+.500*3712.053]
= 0.18699E+06 <= 0.24838E+06  Check Passes
```

Maximum Pressure for Flange on Nozzle H [Pmax]:
```
= min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
= min( 3712.053, 0.500 * 3712.053 + 3712.053 - 186986.6/(pi * 248.539^3) )
= min( 3712.053, 1692.713 )
= 1692.713 kPa
```

Evaluating Flange on Nozzle [J]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(7354)+4*abs(5500)196.088 <= pi*196.088^3[(3712.053-420.361)+.500*3712.053]
= 0.12198E+06 <= 0.12198E+06  Check Passes
```

Maximum Pressure for Flange on Nozzle J [Pmax]:
```
= min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
= min( 3712.053, 0.500 * 3712.053 + 3712.053 - 121978.2/(pi * 196.088^3) )
= min( 3712.053, 420.361 )
= 420.361 kPa
```

Evaluating Flange on Nozzle [K]:
```
= 16*ME + 4*FE*G <= pi*G^3[((PR - PD) + FM * PR)]
= 16(33517)+4*abs(15000)357.251 <= pi*357.251^3[(3712.053-420.361)+.500*3712.053]
= 0.55771E+06 <= 0.73765E+06  Check Passes
```

Maximum Pressure for Flange on Nozzle K [Pmax]:

```
  = min( PR, FM * PR + PR - (16*ME + 4*FE * G) / ( pi * G^3 ))
  = min( 3712.053, 0.500 * 3712.053 + 3712.053 - 557713.1/(pi * 357.251^3) )
  = min( 3712.053, 1676.048 )
  = 1676.048 kPa
```

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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**Input, Nozzle Desc: A From: 30**

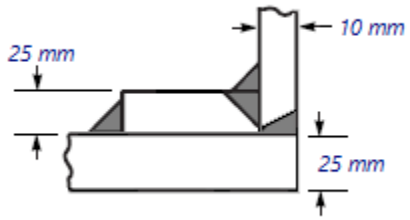
| | | | |
|---|---|---|---|
| Pressure for Reinforcement Calculations | P | 523.703 | kPa |
| Temperature for Internal Pressure | Temp | 360 | °C |
| Design External Pressure | Pext | 101.32 | kPa |
| Temperature for External Pressure | Tempex | 360 | °C |
| Maximum Allowable Pressure New & Cold | | 1110.18 | kPa |
| Parent Material | | PT460 | |
| Parent Allowable Stress at Temperature | Sv | 123008.77 | kPa |
| Parent Allowable Stress At Ambient | Sva | 130990.02 | kPa |
| Inside Diameter of Cylindrical Shell | D | 2111.00 | mm |
| Design Length of Section | L | 3475.9165 | mm |
| Shell Finished (Minimum) Thickness | t | 25.0000 | mm |
| Shell Internal Corrosion Allowance | c | 0.0000 | mm |
| Shell External Corrosion Allowance | co | 0.0000 | mm |
| Distance from Bottom/Left Tangent | | 1980.00 | mm |
| User Entered Minimum Design Metal Temperature | | 0.00 | °C |

***Type of Element Connected to the Parent : Nozzle***

| | | | |
|---|---|---|---|
| Material | | PT460 | |
| Material UNS Number | | SA/AS 15 | |
| Material Specification/Type | | Plate | |
| Allowable Stress at Temperature | Sn | 123008.99 | kPa |
| Allowable Stress At Ambient | Sna | 130990.01 | kPa |
| Diameter Basis (for tr calc only) | | Outside | |
| Layout Angle | | 270.00 | deg |
| Diameter | | 914.4000 | mm |
| Size and Thickness Basis | | Actual | |
| Actual Thickness | tn | 10.0000 | mm |
| Flange Material | | SA-105 | |
| Flange Type | | Weld Neck Flange | |
| Corrosion Allowance | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1 | 1.00 | |
| Joint Efficiency of Nozzle Neck | En | 1.00 | |
| Outside Projection | ho | 189.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell | Wo | 10.0000 | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 5.4864 | mm |
| Pad Material | | PT460 | |
| Pad Allowable Stress at Temperature | Sp | 123008.99 | kPa |
| Pad Allowable Stress At Ambient | Spa | 130990.01 | kPa |
| Diameter of Pad along vessel surface | Dp | 1064.4000 | mm |
| Thickness of Pad | te | 25.0000 | mm |
| Weld leg size between Pad and Shell | Wp | 20.0000 | mm |
| Groove weld depth between Pad and Nozzle | Wgpn | 25.0000 | mm |
| Reinforcing Pad Width | | 75.0000 | mm |
| Flange Class | | 300 | |

```
Flange Grade                                    GR 1.1
Flange Series                                      A
```

The Pressure Design option was Design Pressure + static head.




**Reinforcement CALCULATION, Description: A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

```
Actual Outside Diameter Used in Calculation          914.400  mm.
Actual Thickness Used in Calculation                  10.000  mm
```

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
```
  = P*R/(Sv*E-0.6*P) per UG-27 (c)(1)
  = 524*1056/(123009*1-0.6*524)
  = 4.5052 mm
```

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
```
  = P*R/(Sv*E-0.6*P) per UG-27 (c)(1)
  = 1110*1056/(130990*1-0.6*1110)
  = 8.9914 mm
```

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
```
  = P*Ro/(Sn*E+0.4*P) per Appendix 1-1 (a)(1)
  = 524*457/(123009*1+0.4*524)
  = 1.9432 mm
```

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
```
  = P*Ro/(Sn*E+0.4*P) per Appendix 1-1 (a)(1)
  = 1110*457/(130990*1+0.4*1110)
  = 3.8618 mm
```

Required Nozzle thickness under External Pressure per UG-28 : 1.5996 mm

```
Nozzle Reqd Thickness (Trn) Max(Setup File,Trn)  6.096 mm
```

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**
```
Parallel to Vessel Wall (Diameter Limit)            Dl  1788.7999  mm
Parallel to Vessel Wall, opening length              d   894.4000  mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp    50.0000  mm
```

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Mapnc]**
```
Parallel to Vessel Wall (Diameter Limit)            Dl  1788.7999  mm
Parallel to Vessel Wall, opening length              d   894.4000  mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp    50.0000  mm
```

Weld Strength Reduction Factor [fr1]:
= 1.000

Weld Strength Reduction Factor [fr2]:
= min( 1, Sn/Sv )
= min( 1, 123009/123009 )
= 1.000

Weld Strength Reduction Factor [fr4]:
= min( 1, Sp/Sv )
= min( 1, 123009/123009 )
= 1.000

Weld Strength Reduction Factor [fr3]:
= min( fr2, fr4 )
= min( 1, 1 )
= 1.000

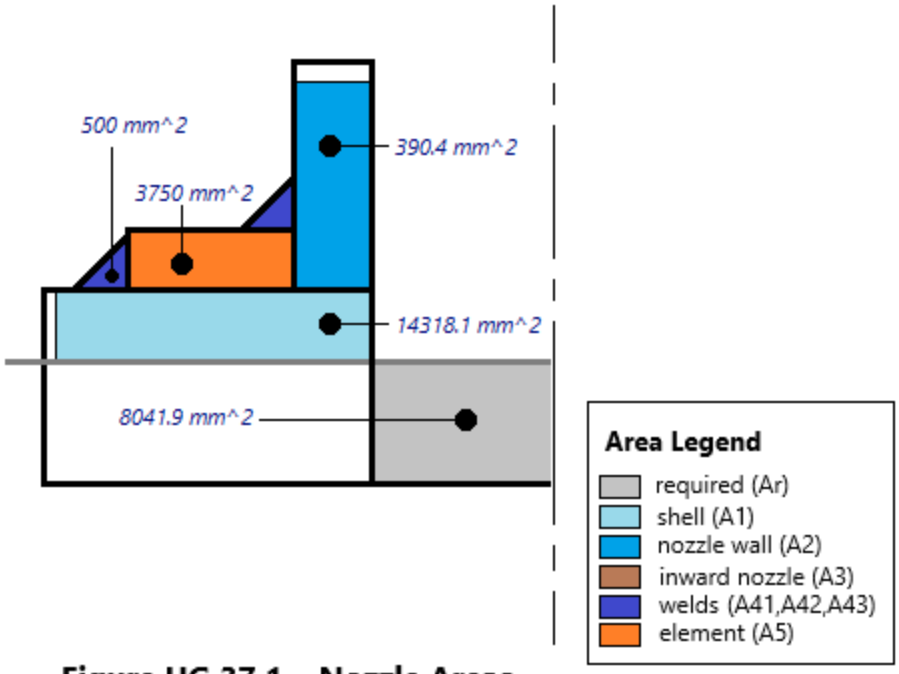



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm^2)**

| AREA AVAILABLE, A1 to A5 | | Design | External | Mapnc |
|---|---|---|---|---|
| Area Required | Ar | 4029.488 | 3990.505 | 8041.898 |
| Area in Shell | A1 | 18330.510 | 14378.987 | 14318.100 |
| Area in Nozzle Wall | A2 | 390.400 | 390.400 | 390.400 |
| Area in Inward Nozzle | A3 | 0.000 | 0.000 | 0.000 |
| Area in Welds | A41+A42+A43 | 500.000 | 500.000 | 500.000 |
| Area in Element | A5 | 3750.001 | 3750.001 | 3750.001 |
| TOTAL AREA AVAILABLE | Atot | 22970.910 | 19019.389 | 18958.500 |

The MAP(nc) Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:
= ( d * tr*F + 2 * tn * tr*F * (1-fr1) ) UG-37(c)
= (894*8.99*1+2*10*8.99*1*(1-1))
= 8041.898 mm^2

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:
= d( E1*t - F*tr ) - 2 * tn( E1*t - F*tr ) * ( 1 - fr1 )
= 894 ( 1 * 25 - 1 * 8.99 ) - 2 * 10
( 1 * 25 - 1 * 8.99 ) * ( 1 - 1 )
= 14318.100 mm^2

Area Available in Nozzle Wall Projecting Outward [A2]:
= ( 2 * Tlwp ) * ( tn - trn ) * fr2
= ( 2 * 50 ) * ( 10 - 6.1 ) * 1
= 390.400 mm^2

Area Available in Welds [A41 + A42 + A43]:
= $Wo^2$ * fr3 + $(Wi-can/0.707)^2$ * fr2 + $Wp^2$ * fr4
= $10^2$ * 1 + $(0)^2$ * 1 + $20^2$ * 1
= 500.000 mm^2

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= ( 1064 - 914 ) * 25 * 1
= 3750.001 mm^2

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**
Wall Thickness for Internal/External pressures ta = 6.0960 mm
Wall Thickness per UG16(b), tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 4.5052 mm
Wall Thickness tb1 = max(trb1, tr16b) = 4.5052 mm
Wall Thickness, shell/head, external pressure trb2 = 0.8699 mm
Wall Thickness tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45 tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[ tb3, max( tb1,tb2) ]
= min[ 8.33, max( 4.51, 1.5 ) ]
= 4.5052 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max( ta, tb )
= max( 6.1, 4.51 )
= 6.0960 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

**UG-45 Minimum Nozzle Neck Thickness Requirement: [MAPnc]**
Wall Thickness for Internal/External pressures ta = 6.0960 mm
Wall Thickness per UG16(b), tr16b = 1.5000 mm

```
 Wall Thickness, shell/head, internal pressure     trb1 = 8.9914 mm
 Wall Thickness                 tb1 = max(trb1, tr16b) = 8.9914 mm
 Wall Thickness, shell/head, external pressure     trb2 = 0.8699 mm
 Wall Thickness                 tb2 = max(trb2, tr16b) = 1.5000 mm
 Wall Thickness per table UG-45                    tb3 = 8.3312 mm
```

Determine Nozzle Thickness candidate [tb]:

```
  = min[ tb3, max( tb1,tb2) ]
  = min[ 8.33, max( 8.99, 1.5 ) ]
  = 8.3312 mm
```

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

```
  = max( ta, tb )
  = max( 6.1, 8.33 )
  = 8.3312 mm
```

Available Nozzle Neck Thickness = 10.0000 mm -> OK

***Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):***

```
 ---------------------------------------------------
 Sustained  :  24855.1,  Allowable : 123009.0 kPa  Passed
 Expansion  :      0.0,  Allowable : 292643.6 kPa  Passed
 Occasional :  11580.5,  Allowable : 163602.0 kPa  Passed
 Shear      :  13232.6,  Allowable :  86106.3 kPa  Passed
```

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

---

Govrn. thk, tg = 10, tr = 1.82, c = 0 mm, E✱ = 1
Thickness Ratio = tr * E✱/(tg - c) = 0.18, Temp. Reduction = 78 °C

```
 Min Metal Temp. w/o impact per UCS-66, Curve A      -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1)  -104 °C
```

**Nozzle Neck to Pad Weld for the Nozzle, Curve: B**

---

Govrn. thk, tg = 10, tr = 1.82, c = 0 mm, E✱ = 1
Thickness Ratio = tr * E✱/(tg - c) = 0.18, Temp. Reduction = 78 °C

```
 Min Metal Temp. w/o impact per UCS-66, Curve B      -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1)  -104 °C
```

**Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B**

---

Govrn. thk, tg = 10, tr = 1.82, c = 0 mm, E✱ = 1
Thickness Ratio = tr * E✱/(tg - c) = 0.18, Temp. Reduction = 78 °C

```
 Min Metal Temp. w/o impact per UCS-66, Curve B      -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1)  -104 °C
```

**Shell to Pad Weld Junction at Pad OD, Curve: B**

---

Govrn. thk, tg = 25, tr = 4.23, c = 0 mm, E✱ = 1

Thickness Ratio = tr * E✱/(tg - c) = 0.17, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            -1 °C
Min Metal Temp. at Required thickness (UCS 66.1)        -104 °C
```

**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B**

---

Govrn. thk, tg = 10, tr = 1.82, c = 0 mm, E✱ = 1
Thickness Ratio = tr * E✱/(tg - c) = 0.18, Temp. Reduction = 78 °C

```
Min Metal Temp. w/o impact per UCS-66, Curve B            -29 °C
Min Metal Temp. at Required thickness (UCS 66.1)        -104 °C

Gov. MDMT of the Nozzle                                :  -104 °C
Gov. MDMT of the Reinforcement Pad                     :  -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :  -104 °C
```

**Flange MDMT including Temperature reduction per UCS-66.1:**

```
Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)      -104 °C
```

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

```
Design Pressure/Ambient Rating = 523.70/5100.00 = 0.103
```

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

```
Parallel to Vessel Wall (Diameter Limit)           Dl  1341.5999  mm
Parallel to Vessel Wall, opening length             d   670.7999  mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp    50.0000  mm
```

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm^2 )**

| AREA AVAILABLE, A1 to A6 | | No Pad | With Pad |
|---|---|---|---|
| Area Required [ MAPnc] | AR | 5361.534 | 5361.534 |
| Area Available in Shell | A1 | 7159.049 | 7159.049 |
| Area Available in Nozzle Wall | A2 | 195.200 | 390.400 |
| Area Available in Inward Nozzle | A3 | 0.000 | 0.000 |
| Area Available in Welds | A4 | 100.000 | 500.000 |
| Area Available in Pad | A5 | 0.000 | 3750.001 |
| Area Available in Hub | A6 | 0.000 | 0.000 |
| TOTAL AREA AVAILABLE | Atot | 7454.249 | 11799.450 |

```
The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.
```

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: A

```
Intermediate Calc. for nozzle/shell Welds   Tmin     10.0000  mm
Intermediate Calc. for pad/shell Welds   TminPad     19.0000  mm
```

**Results Per UW-16.1:**

```
                     Required Thickness    Actual Thickness
Nozzle Weld         6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
Pad Weld            9.5000 = 0.5*TminPad 14.1400 = 0.7 * Wp mm
```

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:
= ( Ar - A1 )* Sv
= ( 8042 - 14318 ) * 123009
= 0.00 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²*fr2)*Sv
= ( 390 + 3750 + 500 - 0 * 1 ) * 123009
= 570794.62 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:
= (pi/2) * Dlo * Wo * 0.49 * Snw
= ( 3.14/2.0 ) * 914 * 10 * 0.49 * 123009
= 865720. N

Shear, Pad Element Weld [Spew]:
= (pi/2) * DP * WP * 0.49 * SEW
= ( 3.14/2.0 ) * 1064 * 20 * 0.49 * 123009
= 2015465. N

Tension, Pad Groove Weld [Tpgw]:
= ( pi/2) * Dlo * Wgpn * 0.74 * Seg
= (3.14/2 ) * 914 * 25 * 0.74 * 123009
= 3268534. N

Shear, Nozzle Groove Weld [Sngw]:
= (pi/2) * (Dlr+Wgnva) * (Wgnva-Can) * 0.60 * Sng
= (3.14/2)*(894 +5.49 )*(5.49 -0 )*0.6*123009
= 572363. N

See UG-41 and UW-15 for clarification of allowable stresses.

**Strength of Failure Paths:**

Failure Path [Path 1-1]:
= ( SPEW + SNGW ) = ( 2015464 + 572363 ) = 2587827 N

**Summary of Failure Path Calculations:**
Path 1-1 = 2587827 N , must exceed W = 0 N or W1 = 570794 N

**Maximum Allowable Pressure for this Nozzle at this Location:**
Converged Maximum Allowable Pressure in the Operating case: 1618 kPa

Nozzle is O.K. for the External Pressure: 101 kPa

Approximate Maximum Allowable Pressure New & Cold: 1825 kPa

The Drop for this Nozzle is : 104.1596 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1596 mm

**Percent Elongation Calculations:**

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.106 %

*Warning:*
*Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).*

**Input Echo, WRC297 Item 1, Description: A :**

| | | | |
|---|---|---|---|
| Diameter Basis for Cylindrical Shell | | ID | |
| Shell Internal Corrosion Allowance | | 0.0000 | mm |
| Shell External Corrosion Allowance | | 0.0000 | mm |
| Shell Diameter | | 2111.000 | mm |
| Shell Thickness | | 25.0000 | mm |
| Shell Stress Concentration Factor | | 1.200 | |
| Vessel Material | | PT460 | |
| Vessel Cold S.I. Allowable | Smc | 130990.02 | kPa |
| Vessel Hot S.I. Allowable | Smh | 123008.77 | kPa |
| Diameter Basis for Nozzle | | OD | |
| Nozzle Corrosion Allowance | | 0.0000 | mm |
| Nozzle Diameter | | 914.400 | mm |
| Nozzle Thickness | | 10.0000 | mm |
| Nozzle Stress Concentration Factor | | 1.000 | |
| Nozzle Material | | PT460 | |
| Nozzle Cold S.I. Allowable | SNmc | 130990.01 | kPa |
| Nozzle Hot S.I. Allowable | SNmh | 123008.99 | kPa |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

| | | |
|---|---|---|
| Diameter of Reinforcing Pad | 1064.400 | mm |
| Thickness of Reinforcing Pad | 25.0000 | mm |

*Note:*
*External Forces and Moments in WRC 107/537 Convention:*
*These loads are assumed to be SUStained loads.*

| | | | |
|---|---|---|---|
| Design Internal Pressure | Dp | 523.70 | kPa |
| Radial Load | P | -25926.99 | N |
| Circumferential Shear | Vc | 22115.01 | N |
| Longitudinal Shear | Vl | 69592.96 | N |
| Circumferential Moment | Mc | -45209.96 | N-m |
| Longitudinal Moment | Ml | 78037.98 | N-m |
| Torsional Moment | Mt | 101000.95 | N-m |
| Include Axial Pressure Thrust | | No | |
| Include Pressure Stress Indices per Div. 2 | | No | |
| Local Loads applied at end of Nozzle/Attachment | | No | |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 * 1.65 * sqrt( Rmean( t - ca ) )
= 914.4 + 2 * 1.65 * sqrt( 1080.5( 25.0 - 0.0 ) )
= 1456.771 mm

WRC 297 Curve Access Parameters:

```
 Vessel Mean Diameter      (D) =   2161.000 mm
 Nozzle Outside Diameter   (d) =    914.400 mm
 Vessel Thickness used     (T) =     50.000 mm
 Nozzle Thickness used     (t) =     10.000 mm
 T / t                         =      5.000
 d / t                         =     91.440

Lambda = [(d/D) * sqrt(D/T)]  = 2.782

 Nr/P    =      0.019
 Mr/P    =      0.019
 M0/P    =      0.043
 N0/P    =      0.229
 MrD/Mc  =      0.059
 NrDL/Mc =      0.049
 M0d/Mc  =      0.166
 N0DL/Mc =      0.503
 MrD/Ml  =      0.031
 NrDL/Ml =      0.027
 M0D/Ml  =      0.060
 N0DL/Ml =      0.528
```

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses Normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Outplane Membrane (P ) | 2847 | 2847 | 2847 | 2847 |
| Outplane Bending (P ) | 3212 | -3212 | 3212 | -3212 |
| Outplane Membrane (Mc) | 0 | 0 | 0 | 0 |
| Outplane Bending (Mc) | 0 | 0 | 0 | 0 |
| Outplane Membrane (ML) | -21600 | -21600 | 21600 | 21600 |
| Outplane Bending (ML) | -14788 | 14788 | 14788 | -14788 |
| Normal Pressure Stress | 10796 | 11320 | 10796 | 11320 |
| Outplane Stress Summary | -19532 | 4143 | 53246 | 17767 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses parallel to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Inplane Membrane (P ) | 227 | 227 | 227 | 227 |
| Inplane Bending (P ) | 1413 | -1413 | 1413 | -1413 |
| Inplane Membrane (Mc) | 0 | 0 | 0 | 0 |
| Inplane Bending (Mc) | 0 | 0 | 0 | 0 |
| Inplane Membrane (ML) | -1082 | -1082 | 1082 | 1082 |
| Inplane Bending (ML) | -7708 | 7708 | 7708 | -7708 |
| Inplane Pressure Stress | 5398 | 5398 | 5398 | 5398 |
| Inplane Stress Summary | -1751 | 10838 | 15830 | -2413 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Shear stress normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Outplane Shear (Vc) | 303 | 303 | -303 | -303 |

| | | | | |
|---|---|---|---|---|
| Outplane Shear (Vl) | 0 | 0 | 0 | 0 |
| Outplane Shear (Mt) | 1530 | 1530 | 1530 | 1530 |
| Shear Stress Summary | 1833 | 1833 | 1227 | 1227 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stress Intensities | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 19718 | 11307 | 53281 | 20325 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Outplane Membrane (P ) | 2847 | 2847 | 2847 | 2847 |
| Outplane Bending (P ) | 3212 | -3212 | 3212 | -3212 |
| Outplane Membrane (Mc) | 11934 | 11934 | -11934 | -11934 |
| Outplane Bending (Mc) | 23682 | -23682 | -23682 | 23682 |
| Outplane Membrane (ML) | 0 | 0 | 0 | 0 |
| Outplane Bending (ML) | 0 | 0 | 0 | 0 |
| Normal Pressure Stress | 5398 | 5398 | 5398 | 5398 |
| Outplane Stress Summary | 47076 | -6715 | -24158 | 16781 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Inplane Membrane (P ) | 227 | 227 | 227 | 227 |
| Inplane Bending (P ) | 1413 | -1413 | 1413 | -1413 |
| Inplane Membrane (Mc) | 1165 | 1165 | -1165 | -1165 |
| Inplane Bending (Mc) | 8425 | -8425 | -8425 | 8425 |
| Inplane Membrane (ML) | 0 | 0 | 0 | 0 |
| Inplane Bending (ML) | 0 | 0 | 0 | 0 |
| Inplane Pressure Stress | 10796 | 11320 | 10796 | 11320 |
| Inplane Stress Summary | 22028 | 2875 | 2847 | 17395 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Outplane Shear (Vc) | 0 | 0 | 0 | 0 |
| Outplane Shear (Vl) | -965 | -965 | 965 | 965 |
| Torsional Shear (Mt) | 1530 | 1530 | 1530 | 1530 |
| Shear Stress Summary | 565 | 565 | 2495 | 2495 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stress Intensities | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 47083 | 9652 | 27461 | 19601 |

***Nozzle Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses in the hoop direction) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Hoop Membrane (P ) | 2371 | 2371 | 2371 | 2371 |
| Hoop Bending (P ) | 0 | 0 | 0 | 0 |
| Hoop Membrane (Mc) | 0 | 0 | 0 | 0 |
| Hoop Bending (Mc) | 0 | 0 | 0 | 0 |
| Hoop Membrane (ML) | -17994 | -17994 | 17994 | 17994 |
| Hoop Bending (ML) | 0 | 0 | 0 | 0 |
| Hoop Pressure Stress | 27792 | 28419 | 27792 | 28419 |
| Hoop Stress Summary | 12168 | 12796 | 48158 | 48786 |

***Nozzle Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses Normal to pipe cross-section) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Axial Membrane (P ) | 910 | 910 | 910 | 910 |
| Axial Bending (P ) | 15147 | -15147 | 15147 | -15147 |
| Axial Membrane (Mc) | 0 | 0 | 0 | 0 |
| Axial Bending (Mc) | 0 | 0 | 0 | 0 |
| Axial Membrane (ML) | -12272 | -12272 | 12272 | 12272 |
| Axial Bending (ML) | -92739 | 92739 | 92739 | -92739 |
| Axial Pressure Stress | 13892 | 13892 | 13892 | 13892 |
| Axial Stress Summary | -75061 | 80122 | 134961 | -80811 |

***Nozzle Stresses (kPa):***

| LONGITUDINAL PLANE (Shear stress) | Au Outside | Al Inside | Bu Outside | Bl Inside |
|---|---|---|---|---|
| Shear due to (Vc) | 1537 | 1537 | -1537 | -1537 |
| Shear due to (Vl) | 0 | 0 | 0 | 0 |
| Shear due to Torsion | 7680 | 7680 | 7680 | 7680 |
| Shear Stress Summary | 9218 | 9218 | 6143 | 6143 |

***Nozzle Stresses (kPa):***

| LONGITUDINAL PLANE (Stress Intensities | Au Outside | Al Inside | Bu Outside | Bl Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 89154 | 81356 | 135389 | 130176 |

***Nozzle Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stresses in the hoop direction) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Hoop Membrane (P ) | 2371 | 2371 | 2371 | 2371 |
| Hoop Bending (P ) | 0 | 0 | 0 | 0 |
| Hoop Membrane (Mc) | 9942 | 9942 | -9942 | -9942 |
| Hoop Bending (Mc) | 0 | 0 | 0 | 0 |
| Hoop Membrane (ML) | 0 | 0 | 0 | 0 |
| Hoop Bending (ML) | 0 | 0 | 0 | 0 |
| Hoop Pressure Stress | 27792 | 28419 | 27792 | 28419 |

| | | | | |
|---|---|---|---|---|
| Hoop Stress Summary | 40105 | 40733 | 20221 | 20849 |

***Nozzle Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Axial Membrane (P ) | 910 | 910 | 910 | 910 |
| Axial Bending (P ) | 15147 | -15147 | 15147 | -15147 |
| Axial Membrane (Mc) | 7108 | 7108 | -7108 | -7108 |
| Axial Bending (Mc) | 102529 | -102529 | -102529 | 102529 |
| Axial Membrane (ML) | 0 | 0 | 0 | 0 |
| Axial Bending (ML) | 0 | 0 | 0 | 0 |
| Axial Pressure Stress | 13892 | 13892 | 13892 | 13892 |
| Axial Stress Summary | 139588 | -95765 | -79687 | 95076 |

***Nozzle Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Shear stress) | Cu Outside | Cl Inside | Du Outside | Dl Inside |
|---|---|---|---|---|
| Shear due to (Vc) | 0 | 0 | 0 | 0 |
| Shear due to (Vl) | -4840 | -4840 | 4840 | 4840 |
| Shear due to Torsion | 7680 | 7680 | 7680 | 7680 |
| Shear Stress Summary | 2840 | 2840 | 12520 | 12520 |

***Nozzle Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stress Intensities | Cu Outside | Cl Inside | Du Outside | Dl Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 139663 | 136616 | 102998 | 97131 |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

```
  = NozzleOD + 2 * 1.65 * sqrt( Rmean( t - ca ) )
  = 914.4 + 2 * 1.65 * sqrt( 1068.0( 25.0 - 0.0 ) )
  = 1453.624 mm
```

WRC 297 Curve Access Parameters:

```
Vessel Mean Diameter      (D) =   2136.000 mm
Nozzle Outside Diameter   (d) =   1064.400 mm
Vessel Thickness used     (T) =     25.000 mm
Nozzle Thickness used     (t) =     85.000 mm
T / t                         =      0.294
d / t                         =     12.522

Lambda = [(d/D) * sqrt(D/T)]  = 4.606
```

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

```
Nr/P    =      0.047
Mr/P    =      0.073
M0/P    =      0.024
N0/P    =      0.064
```

```
MrD/Mc  =      0.250
NrDL/Mc =      0.084
M0d/Mc  =      0.076
N0DL/Mc =      0.086
MrD/Ml  =      0.066
NrDL/Ml =      0.041
M0D/Ml  =      0.021
N0DL/Ml =      0.064
```

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses Normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Outplane Membrane (P ) | 3157 | 3157 | 3157 | 3157 |
| Outplane Bending (P ) | 7211 | -7211 | 7211 | -7211 |
| Outplane Membrane (Mc) | 0 | 0 | 0 | 0 |
| Outplane Bending (Mc) | 0 | 0 | 0 | 0 |
| Outplane Membrane (ML) | -9059 | -9059 | 9059 | 9059 |
| Outplane Bending (ML) | -17312 | 17312 | 17312 | -17312 |
| Normal Pressure Stress | 21848 | 22372 | 21848 | 22372 |
| Outplane Stress Summary | 5846 | 26571 | 58590 | 10066 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stresses parallel to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Inplane Membrane (P ) | 2357 | 2357 | 2357 | 2357 |
| Inplane Bending (P ) | 21904 | -21904 | 21904 | -21904 |
| Inplane Membrane (Mc) | 0 | 0 | 0 | 0 |
| Inplane Bending (Mc) | 0 | 0 | 0 | 0 |
| Inplane Membrane (ML) | -5715 | -5715 | 5715 | 5715 |
| Inplane Bending (ML) | -55570 | 55570 | 55570 | -55570 |
| Inplane Pressure Stress | 10921 | 10921 | 10921 | 10921 |
| Inplane Stress Summary | -26102 | 41229 | 96469 | -58479 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Shear stress normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Outplane Shear (Vc) | 523 | 523 | -523 | -523 |
| Outplane Shear (Vl) | 0 | 0 | 0 | 0 |
| Outplane Shear (Mt) | 2268 | 2268 | 2268 | 2268 |
| Shear Stress Summary | 2792 | 2792 | 1744 | 1744 |

***Vessel Stresses (kPa):***

| LONGITUDINAL PLANE (Stress Intensities) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 32432 | 41739 | 96545 | 68628 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE | Cu | Cl | Du | Dl |
|---|---|---|---|---|

| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
|---|---|---|---|---|
| Outplane Membrane (P ) | 3157 | 3157 | 3157 | 3157 |
| Outplane Bending (P ) | 7211 | -7211 | 7211 | -7211 |
| Outplane Membrane (Mc) | 6991 | 6991 | -6991 | -6991 |
| Outplane Bending (Mc) | 36941 | -36941 | -36941 | 36941 |
| Outplane Membrane (ML) | 0 | 0 | 0 | 0 |
| Outplane Bending (ML) | 0 | 0 | 0 | 0 |
| Normal Pressure Stress | 10921 | 10921 | 10921 | 10921 |
| Outplane Stress Summary | 65222 | -23083 | -22641 | 36817 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Inplane Membrane (P ) | 2357 | 2357 | 2357 | 2357 |
| Inplane Bending (P ) | 21904 | -21904 | 21904 | -21904 |
| Inplane Membrane (Mc) | 6867 | 6867 | -6867 | -6867 |
| Inplane Bending (Mc) | 122434 | -122434 | -122434 | 122434 |
| Inplane Membrane (ML) | 0 | 0 | 0 | 0 |
| Inplane Bending (ML) | 0 | 0 | 0 | 0 |
| Inplane Pressure Stress | 21848 | 22372 | 21848 | 22372 |
| Inplane Stress Summary | 175412 | -112740 | -83190 | 118394 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Outplane Shear (Vc) | 0 | 0 | 0 | 0 |
| Outplane Shear (Vl) | -1661 | -1661 | 1661 | 1661 |
| Torsional Shear (Mt) | 2268 | 2268 | 2268 | 2268 |
| Shear Stress Summary | 606 | 606 | 3929 | 3929 |

***Vessel Stresses (kPa):***

| CIRCUMFERENTIAL PLANE (Stress Intensities) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---|---|---|---|---|
| Two * Max Shear Stress | 175412 | 112740 | 83438 | 118580 |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (kPa):**

| Type of Stress Int. Location | Stress Values at Au | Al | Bu | Bl | Cu | Cl | Du | Dl |
|---|---|---|---|---|---|---|---|---|
| Circ. Pm (SUS) | 10796 | 11320 | 10796 | 11320 | 10796 | 11320 | 10796 | 11320 |
| Circ. Pl (SUS) | -18753 | -18753 | 24447 | 24447 | 1392 | 1392 | -938 | -938 |
| Circ. Q (SUS) | -11576 | 11576 | 18000 | -18000 | 9838 | -9838 | -7012 | 7012 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Long. Pm (SUS) | 5398 | 5398 | 5398 | 5398 | 5398 | 5398 | 5398 | 5398 |
| Long. Pl (SUS) | -855 | -855 | 1309 | 1309 | 14781 | 14781 | -9087 | -9087 |
| Long. Q (SUS) | -6295 | 6295 | 9121 | -9121 | 26894 | -26894 | -20470 | 20470 |
| Shear Pm (SUS) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Shear Pl (SUS) | 303 | 303 | -303 | -303 | -965 | -965 | 965 | 965 |
| Shear Q (SUS) | 1530 | 1530 | 1530 | 1530 | 1530 | 1530 | 1530 | 1530 |
| Pm (SUS) | 10796 | 11320 | 10796 | 11320 | 10796 | 11320 | 10796 | 11320 |
| Pm+Pl (SUS) | 12514 | 11991 | 35246 | 35770 | 20293 | 20301 | 13683 | 14202 |
| Pm+Pl+Q (Total) | 19719 | 11306 | 53283 | 20329 | 47085 | 9655 | 27462 | 19601 |

**Vessel Stress Summation Comparison (kPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---|---|---|---|
| Pm (SUS) | 11320 | 123008 | Passed |
| Pm+Pl (SUS) | 35770 | 184513 | Passed |
| Pm+Pl+Q (TOTAL) | 53283 | 369026 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (kPa):**

| Type of Stress Int. / Location | Stress Values at Au | Al | Bu | Bl | Cu | Cl | Du | Dl |
|---|---|---|---|---|---|---|---|---|
| Circ. Pm (SUS) | 21848 | 22372 | 21848 | 22372 | 21848 | 22372 | 21848 | 22372 |
| Circ. Pl (SUS) | -5902 | -5902 | 12216 | 12216 | 9224 | 9224 | -4510 | -4510 |
| Circ. Q (SUS) | -10101 | 10101 | 24523 | -24523 | 144338 | -144338 | -100530 | 100530 |
| Long. Pm (SUS) | 10921 | 10921 | 10921 | 10921 | 10921 | 10921 | 10921 | 10921 |
| Long. Pl (SUS) | -3358 | -3358 | 8072 | 8072 | 10148 | 10148 | -3834 | -3834 |
| Long. Q (SUS) | -33666 | 33666 | 77474 | -77474 | 44152 | -44152 | -29730 | 29730 |
| Shear Pm (SUS) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Shear Pl (SUS) | 523 | 523 | -523 | -523 | -1661 | -1661 | 1661 | 1661 |
| Shear Q (SUS) | 2268 | 2268 | 2268 | 2268 | 2268 | 2268 | 2268 | 2268 |
| Pm (SUS) | 21848 | 22372 | 21848 | 22372 | 21848 | 22372 | 21848 | 22372 |
| Pm+Pl (SUS) | 15978 | 16500 | 34082 | 34605 | 31340 | 31851 | 17600 | 18112 |
| Pm+Pl+Q (Total) | 32431 | 41742 | 96547 | 68634 | 175413 | 112746 | 83445 | 118580 |

**Vessel Stress Summation Comparison (kPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---|---|---|---|
| Pm (SUS) | 22372 | 123008 | Passed |
| Pm+Pl (SUS) | 34605 | 184513 | Passed |
| Pm+Pl+Q (TOTAL) | 175413 | 369026 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):**

| Type of Stress Int. Location | Stress Values at Au | Al | Bu | Bl | Cu | Cl | Du | Dl |
|---|---|---|---|---|---|---|---|---|
| Circ. Pm (SUS) | 27792 | 28419 | 27792 | 28419 | 27792 | 28419 | 27792 | 28419 |
| Circ. Pl (SUS) | -15623 | -15623 | 20365 | 20365 | 12313 | 12313 | -7571 | -7571 |
| Circ. Q (SUS) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Long. Pm (SUS) | 13892 | 13892 | 13892 | 13892 | 13892 | 13892 | 13892 | 13892 |
| Long. Pl (SUS) | -11362 | -11362 | 13182 | 13182 | 8018 | 8018 | -6198 | -6198 |
| Long. Q (SUS) | -77592 | 77592 | 107886 | -107886 | 117676 | -117676 | -87382 | 87382 |
| Shear Pm (SUS) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Shear Pl (SUS) | 1537 | 1537 | -1537 | -1537 | -4840 | -4840 | 4840 | 4840 |
| Shear Q (SUS) | 7680 | 7680 | 7680 | 7680 | 7680 | 7680 | 7680 | 7680 |
| Pm (SUS) | 27792 | 28419 | 27792 | 28419 | 27792 | 28419 | 27792 | 28419 |
| Pm+Pl (SUS) | 12408 | 13021 | 48268 | 48892 | 41312 | 41903 | 21873 | 22436 |
| Pm+Pl+Q (Total) | 89157 | 81361 | 135392 | 130177 | 139667 | 136616 | 102999 | 97130 |

**Nozzle Stress Summation Comparison (kPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---|---|---|---|
| Pm (SUS) | 28419 | 123008 | Passed |
| Pm+Pl (SUS) | 48892 | 184513 | Passed |
| Pm+Pl+Q (TOTAL) | 139667 | 369027 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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